EXECUTIVE SUMMARY · ADS 503 FINAL PROJECT

Spotting Diabetes Risk

Using everyday health and lifestyle answers to flag who may have diabetes, before costly testing.

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The Problem

Diabetes is common and often goes undiagnosed, yet testing everyone is costly. Could of routine health questions help flag the people most likely to have it?

1 in 7

adults in our data had diabetes or prediabetes.

21 questions

about health and daily habits, with no lab work needed.

Catch it earlier

the goal is to flag high-risk people for follow-up testing.

What We Explored

We analyzed anonymized survey responses, then tested a range of prediction approaches and kept the one that best identified people with diabetes.

70,000+ responses analyzed

An anonymized CDC health survey covering demographics, daily habits, and self-reported health.

Simple to advanced methods

We compared a straightforward model against more advanced pattern-finding models on the same fair test.

Chosen to catch real cases

The winning model was selected for how well it identifies people who truly have diabetes, not just overall accuracy.

What We Found

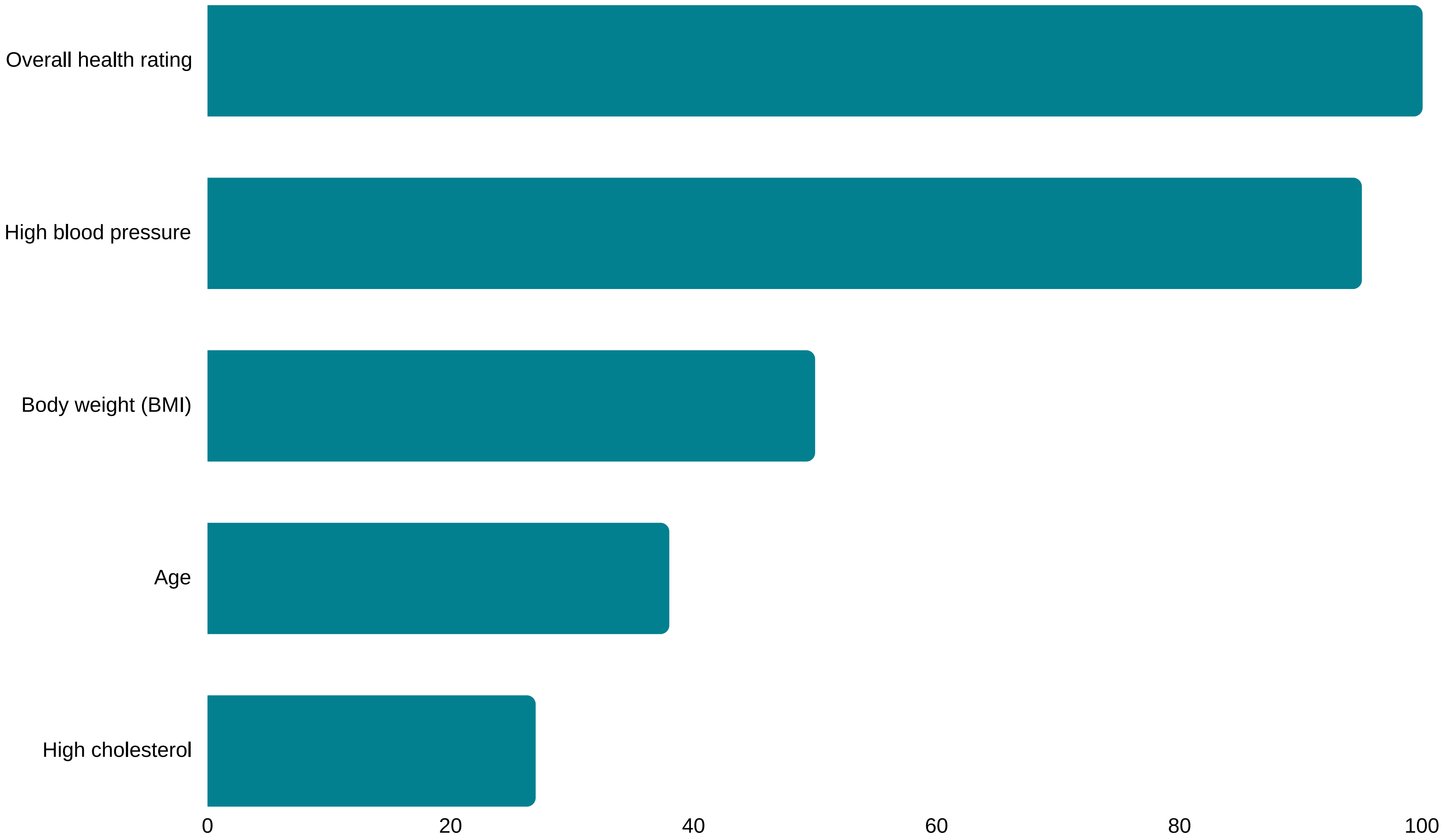
8 in 10

diabetic cases were correctly flagged by the model.

3 in 4

people were classified correctly overall.

What signals risk the most



Relative influence on the model's predictions.

What It Means

- A few everyday factors, mainly overall health, blood pressure, and weight, explain much of the risk.
- A short questionnaire can help target who should be tested, without lab work up front.
- Best used as a screening aid to prioritize follow-up, not as a medical diagnosis on its own.

Next step: validate on the broader population before real-world use.